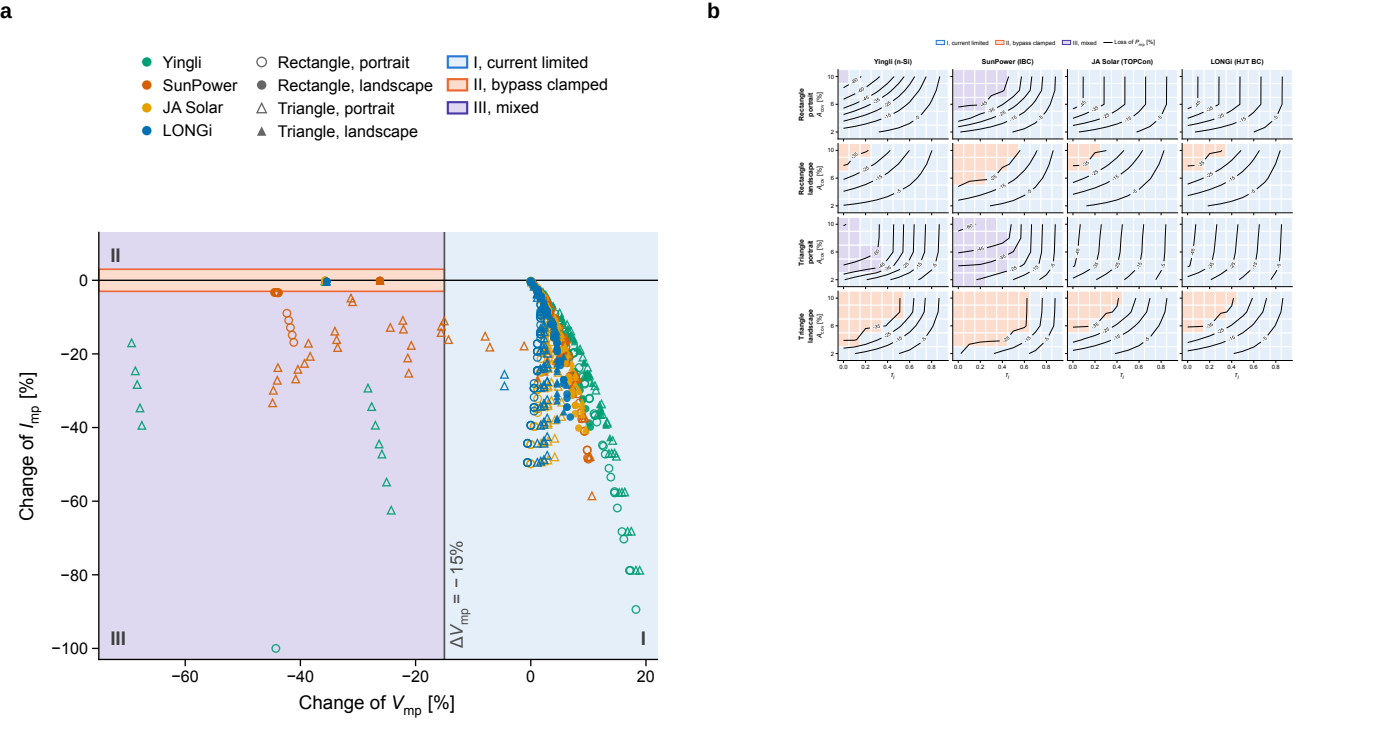

a
Yingli
SunPower
JA Solar
LONGi
Rectangle, portrait
Rectangle, landscape
Triangle, portrait
Triangle, landscape
I, current limited
II, bypass clamped
III, mixed
II
III
I
ΔVmp = −15%
Change of Imp [%]
Change of Vmp [%]
b
I, current limited
II, bypass clamped
III, mixed
Loss of Pmp [%]
Yingli (n-Si)
SunPower (IBC)
JA Solar (TOPCon)
LONGi (HJT BC)
Rectangle portrait
Rectangle landscape
Triangle portrait
Triangle landscape